

Business Discount Rates

Present and Future Value

- Future value (FV) and present value (PV) adjust all cash flows to a common date. This is relevant when we want to compare the cash flows occurring at different periods of time. Either in terms of projects, performance or turnover, the cash flows accrue to the firm at different stages. Evaluation of all these cash flows are true when they are all brought to the same base period.
- In financial parlance, a value of currency is not kept idle. The amount, if invested would certainly bring additional returns in the future. This future expectation from the present investment is termed as the future value.

Relationship between PV and FV

- Assume x amount is invested now and the investor expects r% to accrue on the investment one year ahead. This is translated into present and future values as follows:

$$\text{PV} = \text{INR } x$$

$$\text{FV} = \text{INR } x + (r * x)$$

- Restated as $\text{FV} = \text{PV} * (1+r)$
- This relationship leads to the following concept of discounting the future value to arrive at the present value:

$$\text{PV} = \text{FV} / (1 + r)$$

Compounding of Interest

$$PV = FV / (1 + r)$$

- This is the formula for equating the future value that is associated at the end of 1st year. Now the concept of time over a longer duration can be easily brought into the above equation, where 'n' defines the time duration after which the cash flows are expected.
- If FV is available for two years, its present worth would be defined as
$$PV = FV / (1 + r)^2$$
- Where '2' is the number of years after which the future value is available to a person. Here interest is earned on both the principle and the interest on first year.
- Accumulation of interest as a principle and calculation of interest on this total is called **compounding** of interest.

Interest with/without Compounding

- Interest without compounding is a simple interest formula

$$Pnr/100$$

Where P is the principle, n is the number of years and r is the interest rate.

- Interest with annual compounding adds the interest received earlier to the principle amount and increases the final amount that is received from the investment. Hence, the FV of an investment for a two year duration would be

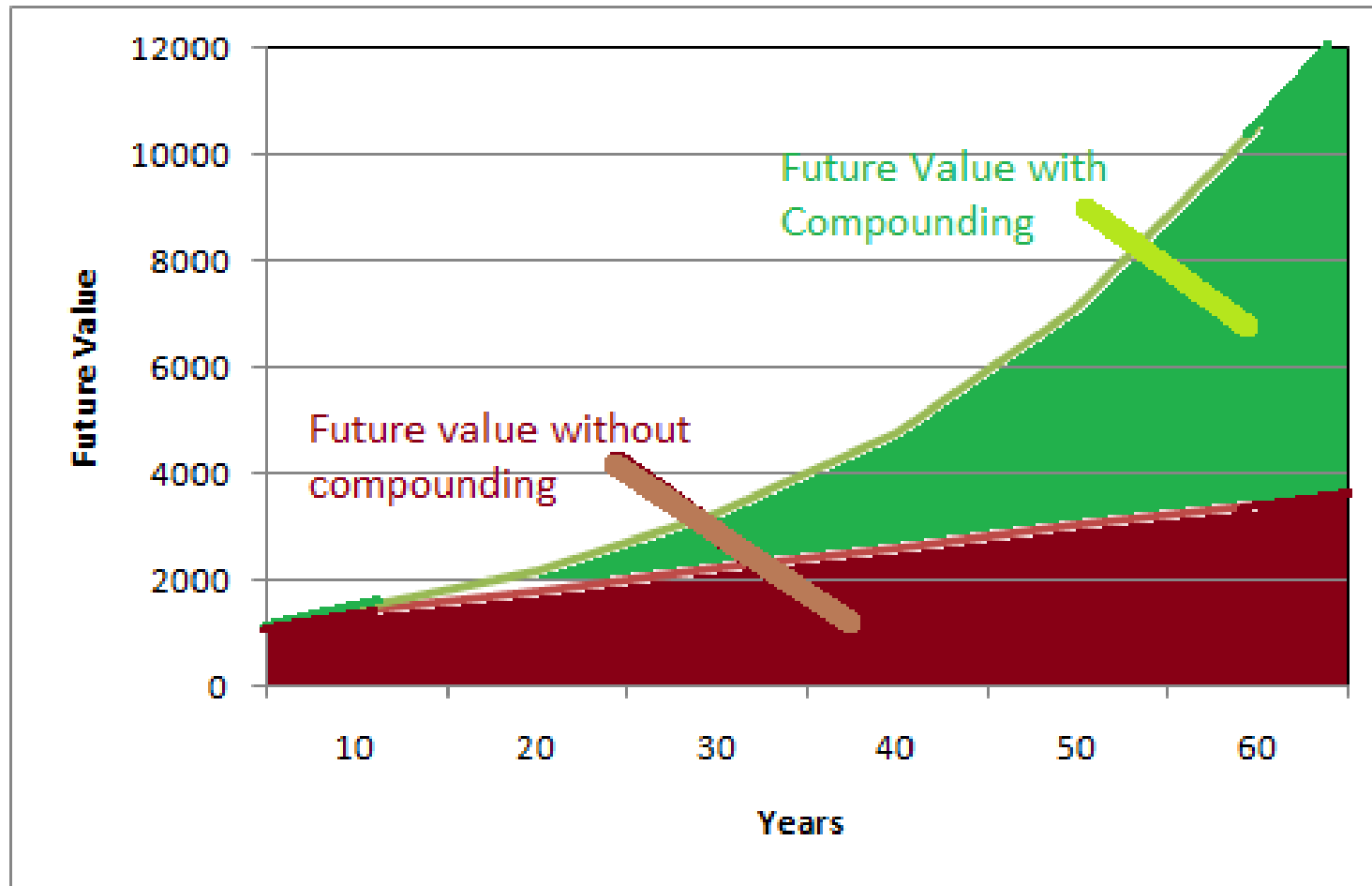
$$FV = PV * (1+r)^* (1+r) = PV * (1+r)^2$$

- Generalized for 'n' years as:

$$PV = FV / (1 + r)^n$$

Impact of Compounding

- Future value of 1000 with/without compounding at 4% interest rate.



Compounding for Less than a Year

FV = Principal + interest

$$F_v = P(1+r)^n$$

- The term $(1+r)^n$ is the compound value factor (CVF) of a lump sum of INR 1, and it always has a value greater than 1 for positive r , indicating that CVF increases as r and n increase.
- Usually, it is common practice to compound the interest on a yearly basis. But, there are instances when compounding is done on a ***half-yearly, quarterly, monthly or a daily basis***. The half-yearly interest rates indicate that interest is payable semiannually, i.e., interest is received $r\%/2$ twice every year. When the principle of compounding is applied, this implies that the $r\%/2$ received twice an year will yield an actual rate which is higher than the declared ($r\%$) rate. This actual rate is called the **effective annual rate**.

$$FV = PV (1 + r/m)^{(m*n)}$$

- where m is the number of times within a year interest is paid.

Multi-period Compounding

- Multi-period compounding is used in practice, when cash flows occur more than once a year.
- Interest rate is usually specified on an annual basis in a loan agreement or security (such as bonds). Annual interest rate is referred as the nominal interest rate. If compounding is done more than once a year the actual rate of interest paid (or received) is the effective interest rate, which is different from the nominal interest rate.
- Effective interest rate would be higher than the nominal interest rate when the compounding is done more frequently.

Effective Interest Rate

- Effective Interest Rate (EIR) is:

$$\text{EIR} = \{[1+r/m]^{(n \times m)}\} - 1$$

- Where r is the annual nominal rate of interest, n number of years and m the number of compounding per year.
- Future Compound Rate

$$F_v = PV\{[(1+r/m)^{(n \times m)}]\}$$

- Where FV is the future value, PV is the present value, r is the annual rate, n is the number of years and m is the number of compounding per year.

Continuous Compounding

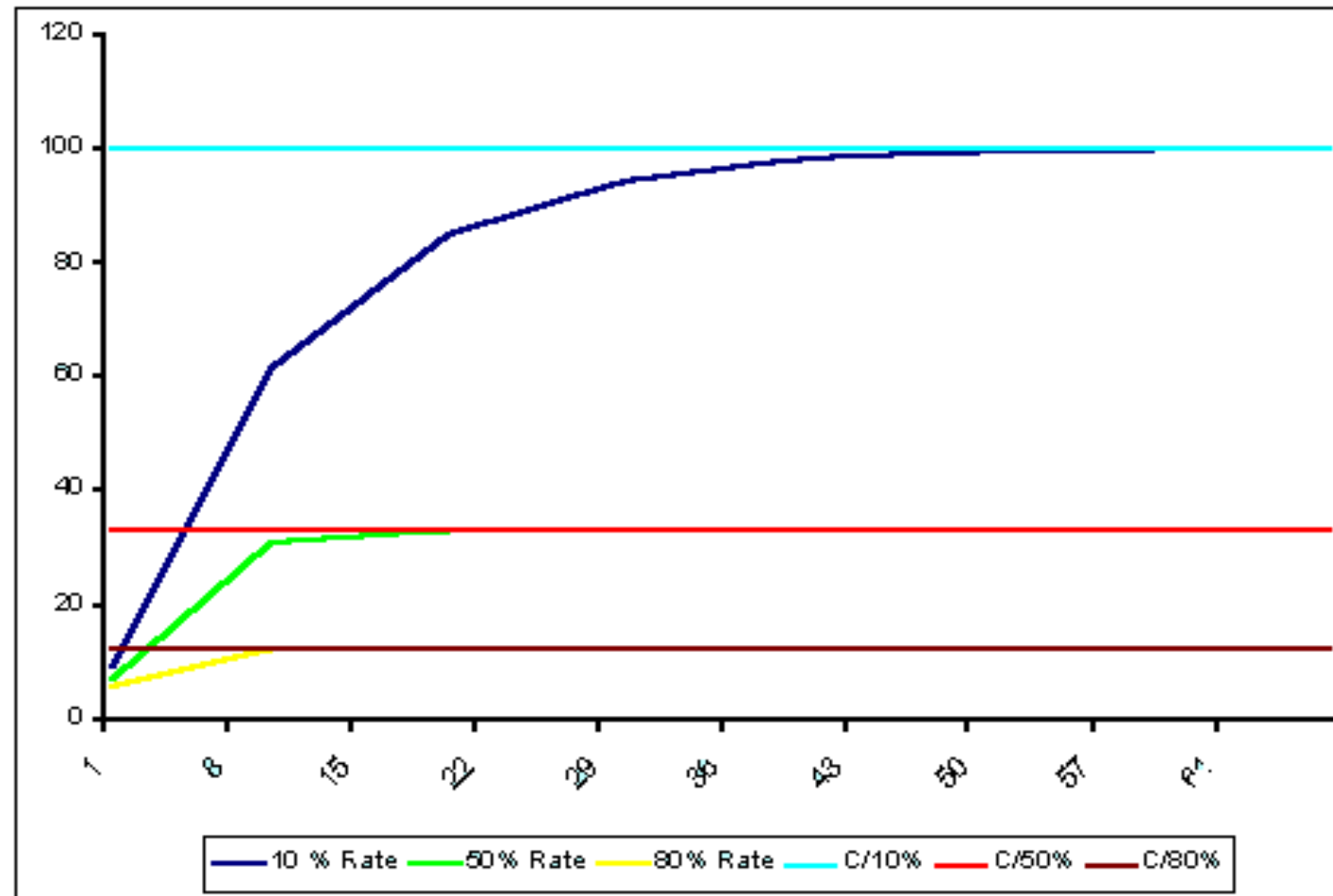
- Sometimes compounding may be done continuously. For example, banks may pay interest continuously; they call it daily compounding. It can be mathematically proved that the continuous compounding function will reduce to the following:

$$FV = P \{e^i\}$$

When $i = (r * n)$ and e is mathematically defined as 2.7183.

Cash Flows

PERPETUAL FLOW



Case: Investment

- Greetez identified an investment opportunity with an expected return of 10 percent.
- Question 1: How much will the firm have in three years if the investment value is INR 400 million?
- Question 2: If the firm can extend its investment project for seven years what will be the future value of investments? At the end of seven years, identify how much interest the firm would have earned? How much of that interest results from compounding?
- Question 3: Greetez will require intermediate development charges that are to be met from the return it generates from the investment, hence, the firm will not be able to reinvest its earnings back into the project. As an alternative what should be the investment rate that would provide the firm with an equal investment value (with reinvestment opportunity) after 7 years?
- Question 4: Greetez requires an additional development expense of INR 200 million for the investment at the end of 3 years. How much should it invest right now to meet the expenditure?

Question 1

- How much will the firm have in three years if the investment value is INR 400 million?
- Investment $\times (1 + r)^3 = 400 \times 1.331 = \text{INR } 532.4 \text{ million}$

Question 2

- If the firm can extend its investment project for seven years what will be the future value of investments? At the end of seven years, identify how much interest the firm would have earned? How much of that interest results from compounding?
- Investment $\times (1 + r)^7$
= 400×1.9487
= INR 779.5 million

Year	Investment	Simple interest	Compound interest	Total interest
1	400.00	40.00	0.00	40.00
2	440.00	40.00	4.00	44.00
3	484.00	40.00	8.40	48.40
4	532.40	40.00	13.24	53.24
5	585.64	40.00	18.56	58.56
6	644.20	40.00	24.42	64.42
7	708.62	40.00	30.86	70.86
Total value				779.49

Question 3

- Greetez will require intermediate development charges that are to be met from the return it generates from the investment, hence, the firm will not be able to reinvest its earnings back into the project. As an alternative what should be the investment rate that would provide the firm with an equal investment value (with reinvestment opportunity) after 7 years?
- Return from the investment = $400 \times 7 \times 0.1 = \text{INR } 280 \text{ million}$
- Investment value at the end of 7 years = $\text{INR } 280 \text{ million} + \text{INR } 400 \text{ million} = \text{INR } 680 \text{ million}$
- Simple interest rate of 13.55% would equate the compound interest rate of 10%.

Question 4

- Greetez requires an additional development expense of INR 200 million for the investment at the end of 3 years. How much should it invest right now to meet the expenditure?
- Additional investment requirement with compounding = 200 million / $(1.1)^3$ = INR 150.26 million

Compound Value of an Annuity

- There can be a uniform cash flow accrual every year over a period of 'n' years. The uniform flow is called "**Annuity**".
- An annuity is a fixed payment (or receipt) each year for a specified number of years. The future compound value of an annuity as follows:

$$FV = A \{[(1+r)^n - 1] / r\}$$

- The term within the curly brackets {} is the ***compound value factor*** for an annuity of Re.1, and A is the annuity.
- Present value of an annuity hence will be

$$PV = A \{[1 - 1/(1+r)^n] / r\}$$

Compound Value of an Annuity

- When an annuity's cash payments are made at the end of each period, it is referred to as regular annuity. On the other hand, the annual payments/receipt can also be made at the beginning of each period. This is referred to as *annuity due*.
- Lease is a contract in which lease rentals (payment) are to be paid for the use of an asset. Hire purchase contract involves regular payments (installments) for acquiring (owning) an asset. A series of fixed payment starting at the beginning of each period for a specified number of periods is called an annuity due.

Compound Value of an Annuity Due

- Compound value of an annuity due is:

$$F_v = A[(1 + r) + (1+r)^2 + (1+r)^3 + \dots + (1+r)^{n-1}]$$

$$FV = A \{[(1+r)^{n-1} - 1] / r\}$$

Hence,

$$PV = A \{[1 - 1/(1+r)^n] / r * (1+r)$$

$$PV = A(PVAR_r)(1+r)$$

Where ***PVAR*** is ***present value of regular annuity*** and r is the interest rate.

Multi Period Annuity Compounding

- Compound value of an annuity in case of the *multi period* compounding is given as follows:

$$FV = A \langle \{ [(1+r/m)^{(n * m)}] - 1 \} / (r/m) \rangle$$

$$PV = A \langle \{ 1 - [1/(1+r/m)^{(n * m)}] \} / (r/m) \rangle$$

- In all instances, the discount rate will be (r/m) and the time horizon will be equal to $(n \times m)$.

Growing Annuity

- An annuity may not be a constant sum through the time duration, it may also grow at a rate of $g\%$ every year. This is referred as a growing annuity. When there is a growth for specific number of years, the present value of an annuity is stated using the following formula:

$$PV = A \left[\frac{1}{r - g} - \frac{1}{r - g} \times \frac{(1 + g)^n}{(1 + r)^n} \right]$$

- Hence, Future value of a growing annuity can be defined by the following formula:

$$FV = A \left[\frac{(1 + r)^n}{(r - g)} - \frac{(1 + g)^n}{(r - g)} \right]$$

Present Value of Perpetuity

- Perpetuity is an annuity that occurs indefinitely. In perpetuity, time period, n , is so large (mathematically n approaches infinity) that the expression $(1+r)^n$ in the present value equation tends to become zero, and the formula for a perpetuity simply condenses into:

$$PV = A/r$$

where A is the annuity amount occurring indefinitely and r is the interest rate.

Present Value of a Growing Annuity

- In financial decision-making there are number of situations where cash flows may grow at a compound rate. Here, the annuity is not a constant amount A but is subject to a growth factor ' g '. When the growth rate ' g ' is constant, the formula can be simplified very easily. The calculation of the present value of a constantly growing perpetuity is given by the following equation:

$$PV = A/(1+r) + A(1+g)/(1+r)^2 + A(1+g)^2/(1+r)^3 + \dots$$

- This equation can be simplified as:

$$PV = A / (r - g)$$

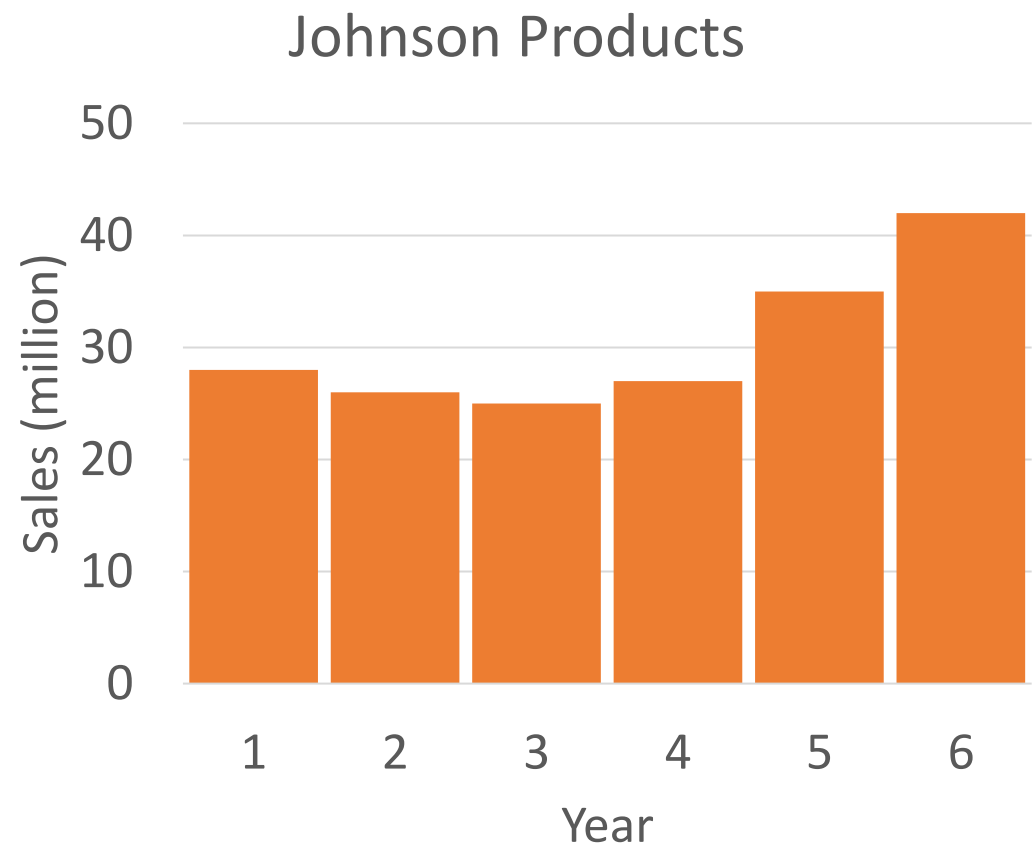
Case Study: Merger Value

- Johnson products a personal care products company proposed to be acquired by a public held company. Icare corp. The proposed acquiring was for about USD **67 million in shares**. Johnson Company's shares soared by USD 5.625 (per share) or 30% on the Stock Exchange, closing at USD 124.625. Shares of Icare, closed unchanged at USD 25.25.
- Most of the two dozen or so major players in the personal-care industry, many of whom need additional capital and marketing capabilities, have viewed consolidation as inevitable, still the potential for fierce competition from the bigger mainstream companies has remained a major concern.

Terms of Merger

- Company's results have improved steadily, however there were instances when it lost USD 2.5 million on reduced sales. Johnson Products, on an average earned USD 3.4 million or USD 2.86 a share on sales of USD 42.9 million.
- Agreement with Icare calls for retaining the same team as chief executive team. Each share of Johnson Products will be converted to the right to receive ten Icare share.
- Icare chairman and chief executive officer, called the merger "Complementary to several of the business Icare is pursuing."

Johnson Products Performance



Case Questions

- 1. Suppose the USD 3.4 million annual income is a perpetuity. What is the present value of Johnson Products' future incomes calculated at a riskless 7% discount rate?
- 2. What is the present value of the acquired firm if we add to the riskless interest rate a 5% risk premium?
- 3. Now assume the income stream is a growing perpetuity beginning at USD 3.4 million and growing at 8% annually. Discount rate is $k=12\%$. What is the present value of the perpetuity?
- 4. The graph in the case shows an income growth rate much greater than 8%. How would your answer to (3) change if the growth rate was 10% for 10 years and 0% thereafter?

Question 1

- Suppose the USD 3.4 million annual income is a perpetuity. What is the present value of Johnson Products' future incomes calculated at a riskless 7% discount rate?
- Applying the perpetuity formula $PV = C/r$, we obtain the PV of all future earnings:

$$PV = \frac{C}{r} = \frac{3.4 \text{ million}}{0.07} = \text{USD } 48.57 \text{ million}$$

Thus according to this formula it would seem that Icare is offering too much.

Question 2

- What is the present value of the acquired firm if we add to the riskless interest rate a 5% risk premium?
- Because the cash flows are uncertain, a risk premium should be added to the discount rate (**7%**). Therefore, with a modest risk premium of **5%** the earnings should be discounted at

$$k = 12\%(7\% + 5\%).$$

- Incorporating uncertainty into the calculation gives us an even lower present value:

$$PV = \frac{C}{r} = \frac{3.4 \text{ million}}{0.12} = \text{USD } 28.33 \text{ million}$$

- Risk factor strengthens our previous conclusion - that Icare's offer is too high.

Question 3

- Now assume the income stream is a growing perpetuity beginning at USD 3.4 million and growing at 8% annually. The discount rate is $k=12\%$. What is the present value of the perpetuity?
- Using the growing perpetuity formula,

$$PV = \frac{C}{k-g} = \frac{3.4 \text{ million}}{0.12-0.08} = \frac{3.672 \text{ million}}{0.04} = \text{USD } 28.33 \text{ million}$$

Question 4

- The graph in the article shows an income growth rate much greater than 8%. How would your answer to (3) change if the growth rate was 10% for 10 years and 0% thereafter?
- Net income was USD 3.4 million. Therefore it will be USD **3.4 million X (1.1) = USD 3.74 million** next year. Present value of these 10 years of income is

$$PV = C \left[\frac{1}{r - g} - \left[\frac{1}{r - g} \times \frac{(1 + g)^T}{(1 + r)^T} \right] \right]$$

$$PV = 3.74 \text{ million} \left[\frac{1}{0.12 - 0.10} - \left[\frac{1}{0.12 - 0.10} \times \frac{(1 + 0.10)^{10}}{(1 + 0.12)^{10}} \right] \right]$$

----- Question 4

- $PV = 3.74 \text{ million} \times [50 - (50 \times 0.8351) \text{ million}] = \text{USD } 3.74 \text{ million} \times 8.245 = \text{USD } 30.84 \text{ million}$
- $\text{USD } 3.4 \text{ million} \times (1.1)^{10} = \text{USD } 8.82 \text{ million}$ in the last year of the 10% growth trend.
- All other cash flows obtained from year 11 on can be included by using the perpetuity formula
- Hence, the PV in terms of year- 10 is: $(8.82 \text{ million} / 0.12) = \text{USD } 73.50 \text{ million}$
- $PV = \frac{73.50}{(1.12)^{10}} = \text{USD } 23.67 \text{ million}$
- Some of these two components gives the PV of Johnson's cash flow
- $PV = \text{USD } 30.84 \text{ million} + \text{USD } 23.67 \text{ million} = \text{USD } 54.51 \text{ million}$
- which is less than what Icare offered.